KERALA DIPLOMA CURRICULUM REVISION 2026

Course 2111 | II | As specified in the official PDF | Source-grounded

Fundamentals of Manufacturing and Tool Engineering

system, solidification and an overview of other casting and moulding methods

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 2111 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 2111.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma Programme
Course code	2111
Course title	Fundamentals of Manufacturing and Tool Engineering
Semester	II

Item	Official information
Course category	As specified in the official PDF
Periods per week	As specified in the official PDF
Periods per semester	As specified in the official PDF
Credits	As specified in the official PDF
Prerequisite	Topic Course code Course Title Semester Basic Physics 2002A Engineering Physics for Mechanical, Structural and Industrial Applications I/II Course Outcome CO Course Outcome Blooms Taxonomy Level CO1 Explain the evolution and fundamentals of Manufacturing and Tool Engineering Understand CO2 Explain sand casting and other basic casting processes along with their applications Understand Explain the basic principle of Metal forming process, Machining and Joining process along with their CO3 Understand application Identify and classify Engineering materials and Tool materials. Describe about heat treatment and CO4 Understand modern trends in Industry CO-PO mapping COURSE ARTICULATION MATRIX PROGRAM OUTCOMES
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- system, solidification and an overview of other casting and moulding methods

Course outcomes

CO	Official outcome	Study level
CO1	3 2 - - - 2 2 - - - 2	See official syllabus
CO2	3 3 2 2 - 2 - - - - 2	See official syllabus
CO3	3 3 2 2 2 2 - - - - 2	See official syllabus
CO4	3 2 2 - 2 3 2 - - - 2 Total 12 10 6 4 4 9 4 - - - 8 Correlation Level 3 2.5 2 2 2 2.25 2 - - - 2 Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - “-” Teaching and Assessment Scheme Examination Scheme Teaching Scheme/Week Self-learning Total Credits Theory Practical (SS)/Week Hours/Semester C Total L T P S CIE ESE Total CIE ESE Total Diploma Curriculum Revision 2026 2111 Page #1 4 0 0 0 6 60 4 40 60 100 0 0 0 100 L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One	See official syllabus

CO	Official outcome	Study level
	unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination Syllabus - Major Topics Instructional Module Title Major Topics Hours L T Evolution and basics of Introduction and classification of manufacturing processes, safety	

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2 - - - 2 2 - - - 2
CO2	CO2 3 3 2 2 - 2 - - - - 2
CO3	CO3 3 3 2 2 2 2 - - - - 2
CO4	CO4 3 2 2 - 2 3 2 - - - 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	manufacturing and Tool 15 0	4	As specified
Module 2	Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0	3	As specified
Module 3	15 0	2	As specified
Module 4	Tool Materials and Modern Trends 15 0	6	As specified

MODULE 1

manufacturing and Tool 15 0

This module is presented from the official syllabus scope for Fundamentals of Manufacturing and Tool Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: manufacturing and Tool 15 0
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

manufacturing and Tool 15 0

manufacturing and Tool 15 0 is an official topic in Module 1 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify manufacturing and Tool 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, manufacturing and tool 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What manufacturing and Tool 15 0 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] manufacturing and Tool 15 0 [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

practices in manufacturing

practices in manufacturing is an official topic in Module 1 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify practices in manufacturing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, practices in manufacturing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What practices in manufacturing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] practices in manufacturing [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Engineering

Engineering is an official topic in Module 1 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Engineering using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, engineering should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Engineering means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Engineering [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Basics of casting, sand casting process, pattern and moulding sand,

Basics of casting, sand casting process, pattern and moulding sand, is an official topic in Module 1 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Basics of casting, sand casting process, pattern and moulding sand, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, basics of casting, sand casting process, pattern and moulding sand, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Basics of casting, sand casting process, pattern and moulding sand, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-□□ Basics of casting, sand casting process, pattern and moulding sand, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
manufacturing and Tool
15            0
                                                 practices in manufacturing
              Engineering
                                                 Basics of casting, sand casting
process, pattern and moulding sand,
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0

This module is presented from the official syllabus scope for Fundamentals of Manufacturing and Tool Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0

Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0 is an official topic in Module 2 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, casting processes gating and risers, casting defects, special casting and moulding 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

processes

processes is an official topic in Module 2 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify processes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, processes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What processes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] processes [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Metal forming, Machining and Metal Forming Process, Machining operations, Press work and Press

Metal forming, Machining and Metal Forming Process, Machining operations, Press work and Press is an official topic in Module 2 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Metal forming, Machining and Metal Forming Process, Machining operations, Press work and Press using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, metal forming, machining and metal forming process, machining operations, press work and press should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Metal forming, Machining and Metal Forming Process, Machining operations, Press work and Press means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Metal forming, Machining and Metal Forming Process, Machining operations, Press work and Press □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Casting Processes                              Gating and Risers, casting defects, special
casting and moulding                 15           0
                                                    processes
                Metal forming, Machining and        Metal Forming Process, Machining
operations, Press work and Press
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

15 0

This module is presented from the official syllabus scope for Fundamentals of Manufacturing and Tool Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: 15 0
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

Press work tool, Metal joining processes

Press work tool, Metal joining processes is an official topic in Module 3 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Press work tool, Metal joining processes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, press work tool, metal joining processes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Press work tool, Metal joining processes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Press work tool, Metal joining processes [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Tool Materials and Heat treatment, Jigs Fixtures and Inspection,

Tool Materials and Heat treatment, Jigs Fixtures and Inspection, is an official topic in Module 3 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Tool Materials and Heat treatment, Jigs Fixtures and Inspection, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, tool materials and heat treatment, jigs fixtures and inspection, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Tool Materials and Heat treatment, Jigs Fixtures and Inspection, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ Tool Materials and Heat treatment, Jigs Fixtures and Inspection, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
15            0
               Press work                           tool, Metal joining processes
                                                    Tool Materials and Heat
treatment, Jigs Fixtures and Inspection,
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Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Tool Materials and Modern Trends 15 0

This module is presented from the official syllabus scope for Fundamentals of Manufacturing and Tool Engineering. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.

- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Tool Materials and Modern Trends 15 0
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

Tool Materials and Modern Trends 15 0

Tool Materials and Modern Trends 15 0 is an official topic in Module 4 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Tool Materials and Modern Trends 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, tool materials and modern trends 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Tool Materials and Modern Trends 15 0 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Tool Materials and Modern Trends 15 0 [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Surface Finishing, Powder Metallurgy, CAD/CAM &CNC

Surface Finishing, Powder Metallurgy, CAD/CAM &CNC is an official topic in Module 4 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Surface Finishing, Powder Metallurgy, CAD/CAM &CNC using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, surface finishing, powder metallurgy, cad/cam &cnc should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Surface Finishing, Powder Metallurgy, CAD/CAM &CNC means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Surface Finishing, Powder Metallurgy, CAD/CAM &CNC [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Detailed Syllabus

Detailed Syllabus is an official topic in Module 4 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Detailed Syllabus using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, detailed syllabus should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Detailed Syllabus means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Detailed Syllabus □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Mode of

Mode of is an official topic in Module 4 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mode of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mode of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Mode of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Mode of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Mapped Instructional

Mapped Instructional is an official topic in Module 4 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mapped Instructional using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mapped instructional should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Mapped Instructional means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Mapped Instructional [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Subtopic Student Learning Outcome delivery Taxonomy Level

Subtopic Student Learning Outcome delivery Taxonomy Level is an official topic in Module 4 of Fundamentals of Manufacturing and Tool Engineering. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Subtopic Student Learning Outcome delivery Taxonomy Level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, subtopic student learning outcome delivery taxonomy level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Subtopic Student Learning Outcome delivery Taxonomy Level means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-□□ Subtopic Student Learning Outcome delivery Taxonomy Level □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Tool Materials and Modern Trends
15            0
                                                   Surface Finishing, Powder
Metallurgy, CAD/CAM &CNC

Detailed Syllabus

Mode of

Mapped                        Instructional
              Subtopic                        Student Learning Outcome
delivery            Taxonomy Level

CO                              Hours

(L/T)
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

- for Self-Learning
- Week Self-Learning description Hours Module I
- Study the evolution of manufacturing (Craft → Mass → Automation). Prepare a comparative chart showing features,
- advantages and limitations of each stage. Refer industrial examples from automotive and aerospace sectors.
- Prepare a classification chart of manufacturing processes (Primary, Secondary, Finishing). Collect real component
- samples/images and identify the manufacturing process used.
- Study safety practices in manufacturing and tool rooms. Prepare a safety checklist for a tool room and identify potential
- hazards with preventive measures.
- Prepare short notes on basic manufacturing terminology and industrial applications. Visit a nearby workshop/tool room (if
- possible) or watch industrial videos and document observations. Module II
- Study the sand casting process in detail. Draw and label a sand mould layout including cope, drag, core, gating and riser. 6
- Prepare a comparison table of different pattern types and pattern materials. Solve numerical problems related to pattern
- allowances (conceptual understanding).
- Study moulding sand composition and properties. Prepare a chart linking properties (permeability, refractoriness, strength)
- with casting quality.
- Analyze common casting defects with causes and remedies. Prepare a defect identification chart with sketches/images.
- Study other casting processes (Die casting, Centrifugal casting, Injection moulding, Blow moulding) and compare with sand 6
- casting. Module III
- Study metal forming processes (Hot vs Cold working). Prepare a comparison chart and list real industrial applications. 6
- Study machining processes (Turning, Milling, Drilling). Draw tool geometry diagrams and identify tool angles. Watch CNC

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
– Theory
                                               CA1                  CA2
CA3                 CA4                     CA5            CA6           ESE

     Mode           Attendance                  Self-learning     Self-learning
Self-learning     Self-learning                  Series      Series           Written
                                                  activities        activities
activities        activities                Exam        Exam                Exam
                                                 (Module 1)        (Module 2)
(Module 3)        (Module 4)              (2 modules) (2 modules)         (theory)
   Duration
2 hours          2 hours       2.5 hours
  Exam mark                                       15                  15
15                    15                  50             50             60
 Converted to
20              20
     Marks                 5                                                  15
20                     60
     Total
40                                                                        60
   Tentative           End of
End of
                                            By 4th week       By 7th week           By
11th week     By 14th week              8th week        15th week
   schedule          semester
semester
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain manufacturing and Tool 15 0. (3 marks)

Answer: Begin with the standard definition or identity of *manufacturing and Tool 15 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain practices in manufacturing. (3 marks)

Answer: Begin with the standard definition or identity of *practices in manufacturing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Engineering. (3 marks)

Answer: Begin with the standard definition or identity of *Engineering*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Basics of casting, sand casting process, pattern and moulding sand,. (3 marks)

Answer: Begin with the standard definition or identity of *Basics of casting, sand casting process, pattern and moulding sand,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0. (3 marks)

Answer: Begin with the standard definition or identity of *Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain processes. (3 marks)

Answer: Begin with the standard definition or identity of *processes*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Metal forming, Machining and Metal Forming Process, Machining operations, Press work and Press. (3 marks)

Answer: Begin with the standard definition or identity of *Metal forming, Machining and Metal Forming Process, Machining operations, Press work and Press*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Press work tool, Metal joining processes. (3 marks)

Answer: Begin with the standard definition or identity of *Press work tool, Metal joining processes*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** ▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ definition, ▯▯▯▯▯▯ principle/steps, ▯▯▯▯▯▯ application ▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯.

Define or explain Tool Materials and Heat treatment, Jigs Fixtures and Inspection,. (3 marks)

Answer: Begin with the standard definition or identity of *Tool Materials and Heat treatment, Jigs Fixtures and Inspection,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** ▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ definition, ▯▯▯▯▯▯ principle/steps, ▯▯▯▯▯▯ application ▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯.

Module 4

Define or explain Tool Materials and Modern Trends 15 0. (3 marks)

Answer: Begin with the standard definition or identity of *Tool Materials and Modern Trends 15 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** ▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ definition, ▯▯▯▯▯▯ principle/steps, ▯▯▯▯▯▯ application ▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯.

Define or explain Surface Finishing, Powder Metallurgy, CAD/CAM &CNC. (3

marks)

Answer: Begin with the standard definition or identity of *Surface Finishing, Powder Metallurgy, CAD/CAM &CNC*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Detailed Syllabus. (3 marks)

Answer: Begin with the standard definition or identity of *Detailed Syllabus*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Mode of. (3 marks)

Answer: Begin with the standard definition or identity of *Mode of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **manufacturing and Tool 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Casting Processes Gating and Risers, casting defects, special casting and moulding 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Press work tool, Metal joining processes**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Tool Materials and Modern Trends 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- URL Modules Covered
- IIT Madras Manufacturing Process Lectures (YouTube/NPTEL) Casting, Forming, Machining

IN-PAGE SEARCH

Search this lesson